# Anthony Lapsansky

Salish Sea Research Center
Northwest Indian College
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## EDUCATION & RESEARCH

2024-present Senior Researcher, Northwest Indian College, Bellingham, WA
**Research Topics**: mechanisms of hybrid air-water locomotion, carryover effects of biotoxins on aquatic invertebrates, sensory ecology and guidance of aquatic invertebrates

2021-2024 Postdoctoral Fellow, University of British Columbia, Vancouver, BC
Advisors: Douglas Altshuler, Douglas Wylie
**Research Topic**: Visual guidance and neural control of avian flight

- Developed a head-mounted, high-speed camera and motion-sensing system for use on free-ranging birds using microelectronics and Python.
- Leveraged custom equipment to investigate how free-flying pigeons use visual motion for guidance through eye, head, and body movements.
- Used high-speed videography and programming in Python, MATLAB, and R to analyze the oculomotor behavior of Anna's hummingbirds (*Calypte anna*).

2016-2021 Ph.D., University of Montana, Missoula, MT
Advisor: Bret Tobalske
**Dissertation Title**: Aquatic locomotion in birds – biomechanics, morphometrics, and evolution

- Applied 3D kinematic analysis to test the prediction that seabirds like alcids (e.g., puffins, murres) are inefficient when moving in air and water relative to single fluid specialists.
- Compiled an unparalleled database of avian wing shape (over 2,400 wings & 900 species) and used these data to test long-held hypotheses about the functional morphology of diving birds. I also made these data freely available (https://bit.ly/diveBirdData).
- Collaborated with the public to describe previously unknown aquatic habits in birds.
- Utilized expertise in biomechanics to assist in studying performance of endangered *Lednia* stoneflies reared at thermal conditions predicted under climate change.

2012-2016 Bachelor of Science, Gonzaga University, Spokane, WA
Advisors: Joseph Haydock, Peter Pauw
Biology with Research Concentration, *magna cum laude*

- Utilized PCR amplification of microsatellites to study parentage in cooperative breeding acorn woodpeckers (*Melanerpes formicivorus)*, with Joseph Haydock.
- Worked towards a method for culturing Hydra using cell culture techniques, with Peter Pauw.

## TEACHING EXPERIENCE

2023 Volunteer Instructor, Northwest Indian College: Salish Sea Research Center Summer Internship

2021 Teaching Assistant, University of Montana, Ornithology

2021 Guest Lecturer, University of Montana, Ornithology

2017-2020 Teaching Assistant, University of Montana, Freshwater Ecology

2018-2020 Teaching Assistant, University of Montana, Comparative Anatomy

2017 Teaching Assistant, University of Montana, Principles of Living Systems

2017 Teaching Assistant, University of Montana, Discover Biology

2015-2016 Teaching Assistant, Gonzaga University, Organic Chemistry

2013-2015 Academic Tutor, Gonzaga University, Athletic Department

## OTHER RELEVANT EXPERIENCE

2013-2016 Professional Falconer, Airstrike Bird Control
- Utilized trained falcons to protect blueberry crops in Whatcom County from damage by invasive pests (European starling, *Sturnus vulgaris*).

2013 Biological Field Technician, Trinity Western University
- Quantified crop damage in Washington State cherry farms to assess potential benefits of habitat enrichment for native predators, the American kestrel (*Falco sparverius*).

2008-2012 Volunteer Fire Fighter, Sandy Point Fire Department
- Assisted with trainings and emergency responses by the Sandy Point Fire Department across the Sandy Point and Sandy Point Heights neighborhoods.

## PUBLICATIONS

* denotes undergraduate mentee co-author

**Lapsansky, A. B.**, Wylie, D. R. and Altshuler, D. L. (2026). Pigeons make slow, divergent eye movements during flight and large, convergent eye movements when landing. *Current Biology* **36**, 1–10.

**Lapsansky, A. B.,** Kreyenmeier, P., Spering, M. and Altshuler, D. L**.** (2025). Hummingbirds use compensatory eye movements to stabilize both rotational and translational visual motion. *Proceedings of the Royal Society B: Biological Sciences* 291, 1–12.

Altshuler, D. L., Baliga, V. B., **Lapsansky, A. B**., Lee, P., Press, E. R. and Theriault, J. S. (2025). Understanding mechanisms of avian flight by integrating observations with tests of competing hypotheses. *J Exp Biol* **228**, JEB247992.

Dash, S., Baliga, V. B., **Lapsansky, A. B.**, Wylie, D. R. and Altshuler, D. L. (2024). Encoding of global visual motion in the avian pretectum shifts from a bias for temporal-to-nasal selectivity to omnidirectional excitation across speeds. *eNeuro*.

Shah A, Hotaling S, **Lapsansky AB**, Malison R, Birrell J, Keeley T*, Giersch J, Tronstad L, Woods A. Warming undermines emergence success in a threatened alpine stonefly: A multi-trait perspective on vulnerability to climate change. *Functional Ecology.* 37, 1033-1043 (2023).

**Lapsansky AB,** Armstrong RH. Common Mergansers (*Mergus merganser*) use wings to pursue a fish underwater. *Marine Ornithology*. 50, 111–114 (2022).

**Lapsansky AB**, Warrick DR, Tobalske BW. High wing-loading correlates with dive performance in birds, suggesting a strategy to reduce buoyancy. *Integrative and Comparative Biology.* 62, 878–889 (2022).

**Lapsansky AB**, Zatz D, Tobalske BW. Alcids 'fly' at efficient Strouhal numbers in both air and water but vary stroke velocity and angle. *eLife.* 9, e55774 (2020).

**Lapsansky AB** & Tobalske BW. Upstroke-based acceleration and head stabilization are the norm for the wing-propelled swimming of alcid seabirds. *J Exp Biol.* 222, jeb201285 (2019).

**Lapsansky AB**, Igoe JA*, Tobalske BW. Zebra finch (*Taeniopygia guttata*) shift toward aerodynamically efficient flight kinematics in response to an artificial load. *Biology Open.* 8, bio042572 (2019).

## INVITED PRESENTATIONS

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| 2023 | University of British Columbia, Evening Comparative Physiology Seminar, Vancouver, BC |
| 2022 | University of British Columbia, Biodiversity Research Seminar, Vancouver, BC |
| 2022 | Society for Integrative and Comparative Biology (SICB) Symposium, *Lesser known transitions: organismal form and function across abiotic gradients*, Phoenix, AZ |
| 2019 | Gonzaga University, Biology Seminar Series, Spokane, WA |

## GRANTS AND FELLOWSHIPS Total (USD): $649,550

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| 2025 | USDA NIIFA New Discovery Grant | $350,000 |
| 2023 | Michael Smith Health Research BC, Postdoctoral Fellowship | $145,000 |
| 2021 | National Science Foundation, Postdoctoral Fellowship | $138,000 |
| 2021 | University of Montana, Bertha Morton Fellowship | $3,000 |
| 2019 | Drollinger-Dial Foundation, Travel Award | $1,500 |
| 2018 | University of Washington, Stephen and Ruth Wainwright Endowed Fellowship | $900 |
| 2018 | Drollinger-Dial Foundation, Travel Award | $1,500 |
| 2017 | Montana Space Grant Consortium, Research Fellowship | $9,650 |

## HONORS AND AWARDS

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| 2021 | Society for Integrative and Comparative Biology (SICB) Division of Biomechanics, Best Student Presentation |
| 2021 | SICB Division of Phylogenetics and Comparative Biology, David and Marvalee Wake Award for Best Student Presentation |
| 2017 | National Science Foundation, GRFP Honorable Mention |
| 2016 | Gonzaga University, McDonald Award for Academic Distinction |

## STUDENT MENTORSHIP * denotes co-authored publication

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| 2022 | Alonso Daboub, spatial position of optic flow and visual guidance of flight in zebra finches |
| 2022 | Emily Saysanasy, Minda Zhang, Janice Wong, and Dhivya Thiagarajan, *BRC Biodiversity Undergraduates in Research Program* |
| 2018 | Tylor Keeley*, surface skimming performance of stoneflies reared at variable temperatures |
| 2016 | Jennifer Igoe*, kinematic responses of zebra finches to artificial loading |

## COMMUNITY OUTREACH

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| 2023 | Volunteer Instructor, Northwest Indian College: Salish Sea Research Center Summer Internship (8-weeks, 1-2 days per week of instruction and mentorship) |
| 2019-2017 | Lab Docent, Montana Natural History Center: Wings Over Water Program |

2016 Presentation, Vista Middle School, *Raptor Biology and Falconry*, Ferndale, WA

2016 Volunteer Instructor, Gonzaga University, *Science in Action* with Holmes Elementary School (8 hands-on lessons in science)

2015 Presentation, Ferndale High School, *Raptors, Falconry and Bird Abatement: Turning a Passion into a Career*. Ferndale, WA

2015 Presentation, Vista Middle School. *Raptors, Falconry and Bird Abatement.* Ferndale, WA

2014 Presentation, Barnyard Kids 4H group, *Falcons and other natural predators: the importance of wildlife to healthy agriculture.* Lynden, WA

## PROFESSIONAL DEVELOPMENT

2024 University of British Columbia, Equity, Diversity and Inclusion Learning Journey (online course)

2020 University of Washington, 3D Morphometrics and Image Analysis Intense Winter Workshop

2019 University of Montana, Professional Scientist Skills (semester-long course)

2018 University of Montana, Evidence-Based Teaching in Science (semester-long course)

2017 University of Montana, Diane Ebert-May Scientific Teaching Workshop

2016 University of Montana, Grant Proposal Writing (semester-long course)

## SERVICE

**Reviewer**: *Proceedings of the Royal Society B (2)*, *Ecological and Evolutionary Physiology* (1), *Proceedings of the Royal Society Interface (1), Nature Scientific Reports (1)*, *Animal Behavior (1)*, *Polar Biology (1)*, *Journal of Experimental Biology*, *Journal of Avian Biology (1), Biological Journal of the Linnean Society (2)*

Integrative and Comparative Biology, Assistant Editor, (2025-present)

University of Montana, Co-President of Graduate Students in Ecology and Evolution (2020-2021)

Society for Integrative and Comparative Biology, Session chairperson (2017-2021)

University of Montana, Seminar Series Organizer (2018-2020)